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# A systematic review and meta-analysis of randomized controlled trials on the effectiveness of immersive virtual reality in cancer patients receiving chemotherapy

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ABSTRACT

*Purpose:* Immersive virtual reality (IVR) shows promise in cancer care, especially for chemotherapy patients. This systematic review and meta-analysis assesses IVR's impact on adult and pediatric cancer patients undergoing chemotherapy.
*Methods:* We searched PubMed, Cochrane Library, Embase, Scopus, Web of Science, and Google Scholar for relevant randomized controlled trials (RCTs). We focused on anxiety, depression, fatigue, pain, and anxiety in adults and pain and anxiety in pediatric patients.
*Results:* Fifteen trials were included, enrolling 607 adult and 257 pediatric cancer patients. IVR significantly reduced anxiety (SMD = −1.89, 95% CI = −2.93 to −0.85), depression (SMD = −1.85, 95% CI = −3.14 to −0.55), fatigue (SMD = −3.40, 95% CI = −5.54 to −1.26), and systolic blood pressure (MD = −3.54, 95% CI = −6.67 to −0.40) in adults. In pediatric patients, IVR significantly lowered pain (SMD = −1.17, 95% CI = −1.84 to −0.50) and anxiety (SMD = −1.18, 95% CI = −1.77 to −0.59) but not heart rate (MD = 0.48, 95% CI = −2.38 to 3.34).
*Conclusion:* IVR effectively reduces anxiety, depression, fatigue, systolic blood pressure, pain, and anxiety in cancer patients. More robust RCTs are needed for further IVR research.

## 1. Introduction

Chemotherapy is currently among the most reliable cancer treatment modalities for treating various types of malignancies (Ismail et al., 2020; Siegel et al., 2012). Globally, the number of patients who receive chemotherapy as the first modality is expected to increase from 9.8 million in 2018 to 15 million in 2040, an increase of 53% (Wilson et al., 2019). However, despite the benefits of chemotherapy, it exerts a negative effect on the physical (e.g., pain, fatigue, nausea, alopecia, and sexual dysfunction), psychological (e.g., anxiety, distress, and depressive symptoms), and social well-being of patients either due to side effects, the treatment environment, or the burden of the chemotherapy regimen (Battisti et al., 2021; Magalhães et al., 2020; Pearce et al., 2017). Therefore, the current standard of chemotherapy should be strengthened to achieve expected health outcomes either through the implementation of therapeutic management strategies or innovation of alternative chemotherapy methods (Rutkowski et al., 2021).

Immersive Virtual reality (IVR) is a type of VR that aims to completely immerse users in a computer- or device-generated world, enabling users to believe that they have stepped into the virtual world (Liberatore and Wagner, 2021). VR is a three-dimensional virtual environment generated using technological products (such as computers and smartphones). VR has been applied in health and medical fields to achieve the best treatment outcomes for patients (Javaid and Haleem, 2020; Mazurek et al., 2019). VR offers an immersive experience, which can distract individuals from an unpleasant stimulus by affecting the sensory system, causing difficulty focusing on stimuli outside the area of attention (Sikka et al., 2019; Spiegel et al., 2019). VR can provide visual and auditory stimuli in a virtual environment, thus exerting a therapeutic effect on patients (Chow et al., 2021; Javaid and Haleem, 2020).

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VR is essential as a learning support tool in nurse training (Wang et al., 2022a). It is also considered an alternative therapy because it overcomes some of the limitations of conventional treatment (Morina et al., 2015). VR technology is still in its early stages. Still, it is applicable and has the potential to become a non-pharmacological therapy to manage the symptoms of patients with cancer caused by invasive procedures and treatments in the future (Ioannou et al., 2022; Mazurek et al., 2019). VR is emerging as a promising, noninvasive, and easily tolerated tool for treating pediatric and adult cancer patients (Chirico et al., 2016; Huang et al., 2022; Morina et al., 2015). Chemotherapy is an area of cancer treatment in which VR can be used (Chirico et al., 2016). Patients with cancer experience various symptoms before, during, and after chemotherapy. Thus, VR can be applied to reduce physical and psychological symptoms (e.g., anxiety, pain, distress, and fatigue) (Chirico et al., 2020; Rutkowski et al., 2021).

Several nonrandomized studies on the implementation of VR in chemotherapy have demonstrated that VR effectively reduced anxiety, pain, fatigue, and stress and provided a positive experience for chemotherapy patients (Scates et al., 2020; Schneider et al., 2004; Wilson and Scorsone, 2021). Furthermore, recent randomized controlled trials (RCTs) have reported promising results regarding the benefits of VR. Studies have shown that VR effectively distracted patients with breast and ovarian cancer patients during chemotherapy, thus alleviating psychological stress, anxiety, depression, and fatigue (Fabi et al., 2022; Zhang et al., 2022b). However, patients did not experience an improvement in nausea and pain.

A limited number of meta-analysis studies have focused on using virtual reality (VR) in cancer patients, both in adult and pediatric populations, during standard oncologic treatment. In the adult population, a recent study by Rutkowski et al. (2021) conducted a meta-analysis of six trials and found no significant differences between the control and VR groups in terms of anxiety and fatigue (Rutkowski et al., 2021). In contrast, a meta-analysis by Burrai et al. (2023) of eight trials showed that VR significantly reduced anxiety, but did not affect pain and fatigue (Burrai et al., 2023). These studies examined similar outcomes but yielded inconsistent and inconclusive findings. In the pediatric population, a recent meta-analysis by Czech et al. (2023) of nine trials found that VR significantly reduced pain and anxiety (Czech et al., 2023). However, this is the only study focused on pediatric patients, so there are no comparative findings. It should be noted that these studies still focused on patient-reported outcomes (PROs), but physiological indicators such as blood pressure and heart rate have yet to be evaluated. PROs, such as pain and anxiety, are important in the evaluation of VR, but the assessment of bio-parameters is also crucial in the implementation of VR in stressful situations for cancer patients (Ioannou et al., 2022; Ahmad et al., 2020).

Therefore, examining the current evidence on the application of immersive virtual reality (IVR) in chemotherapy settings and strengthening the evidence is crucial to support standard care for cancer patients. This study investigated the effects of IVR on psychological aspects (anxiety, distress, depression, and distress), physical aspects (fatigue and pain), and objective measurements (blood pressure and heart rate) in cancer patients receiving chemotherapy. The primary outcomes of this study were psychological, while secondary outcomes included physical outcomes and objective measurements.

## 2. Methods

### 2.1. Study design

This systematic review was conducted in accordance with the Cochrane Handbook for Systematic Reviews of Interventions and Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). The study protocol was prospectively registered in PROSPERO (registration number: CRD42022359886).

### 2.2. Search strategy

We searched for relevant studies in the following electronic databases: PubMed, Cochrane Library, Embase, Scopus, Web of Science, and MEDLINE. Additionally, we identified studies using the Google Scholar academic search engine and manually identified related studies in the reference lists of selected articles. We used these methods to identify articles that were retrieved during the search but not found during the screening process because of the absence of direct relevance shown in bibliographic records when, in reality, the articles had similarities (Janssens and Gwinn, 2015; Wright et al., 2014). The initial search was conducted in July 2023. We used the following keywords during our search: immersive" OR "virtual reality" OR "head-mounted display" OR "smart glasses" AND "chemotherapy". These keywords were based on medical subject headings. Studies were screened on the basis of inclusion and exclusion criteria, and no language restriction was applied. In addition, we contacted the corresponding authors of eligible articles to request missing data. Fig. 1 presents the PRISMA flowchart.

### 2.3. Eligibility criteria

The inclusion criteria followed the PICO framework (patient/problem, intervention/exposure, comparison/control, and outcome). We included studies that (a) enrolled cancer patients receiving chemotherapy (regardless of age); (b) used IVR in the experimental group; (c) were RCTs; and (d) examined improvement in psychological symptoms (anxiety, distress, and depression), physical symptoms (pain and fatigue), and objective measurements as the outcome of interest. We excluded trials that did not use an IVR environment. Furthermore, duplicate studies were removed using Endnote software (Endnote 20, Clarivate, Version 20.2, London, UK, 2021). Two reviewers independently screened the titles and abstracts of the studies on the basis of accessibility criteria. Any disagreement was discussed until a consensus was reached with a third reviewer.

### 2.4. Data extraction

Two reviewers extracted the following data from included trials: name of the first author, year of publication, inclusion criteria, study design, sample size, patients' age, trial setting, characteristic intervention (consisting of the VR device, virtual environment, duration, and frequency).

### 2.5. Risk of bias assessment

We used a revised Cochrane risk-of-bias tool for randomized trials (Sterne et al., 2019) for risk-of-bias assessments. The methodological quality was determined on the basis of the following bias categories: bias arising from the randomization process, bias due to deviations from intended interventions, bias due to missing outcome data, bias in the measurement of the outcome, and bias in the selection of reported results. The overall risk of bias for all the domains was classified as "high," "some concerns," and "low."

### 2.6. Data analysis

Statistical analysis was performed using Review Manager, version 5.4 (The Nordic Cochrane Center, The Cochrane Collaboration, Copenhagen, Denmark). Heterogeneity was examined using forest plots that used both Q (a significant result indicating statistical heterogeneity) and $I^2$ statistics to measure statistical heterogeneity among trials. Values of 25%, 50%, and 75% indicate low, moderate, and high heterogeneity, respectively (Higgins et al., 2003). A random-effects model was used to calculate pooled results if $I^2 \geq 50\%$; otherwise, a fixed-effects model was used. Forest plots were generated to present the pooled effect. Standardized mean difference (SMD) and mean difference (MD) with the

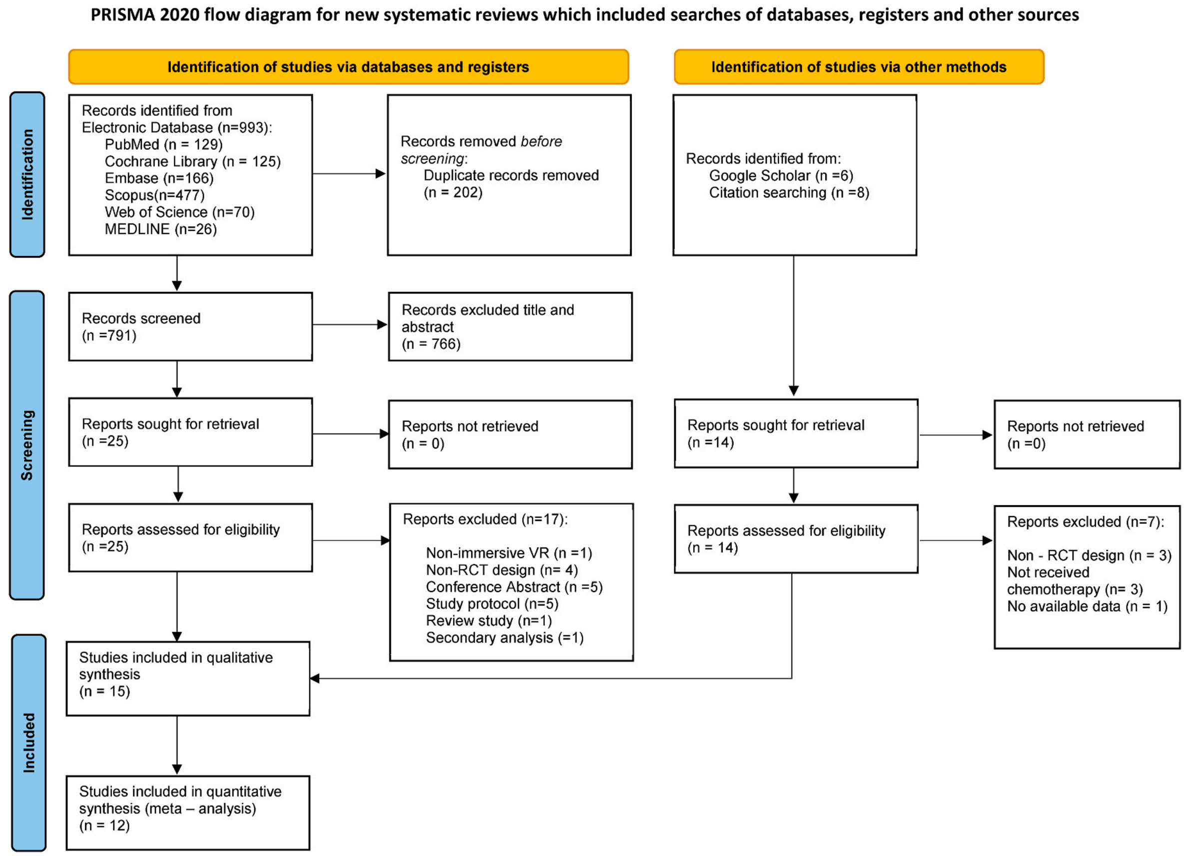


**Fig. 1.** PRISMA Flow Diagram of Study Selection Process
*Consider, if feasible to do so, reporting the number of records identified from each database or register searched (rather than the total number across all databases/ registers).
*If automation tools were used, indicate how many records were excluded by a human and how many were excluded by automation tools.
*From:* Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021; 372:n71. https://doi.org/10.1136/bmj.n71. For more information, visit: http://www.prisma-statement.org/.

corresponding 95% confidence interval (CI) were used to calculate the effect size. The SMD was calculated when the same outcome was reported but using different measurement tools. All tests were two-sided, and a p-value of $<0.05$ was considered statistically significant.

### 2.7. GRADE quality of evidence

The Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) system was used to assess the quality of evidence for each outcome by considering factors such as study risk of bias, publication bias, imprecision, and inconsistency (Guyatt et al., 2008). Based on these assessments, the GRADE system assigns grading levels of high, moderate, low, or very low to indicate the quality of evidence for each outcome. The assessment of the quality of evidence in our findings focused on the outcomes of anxiety, fatigue, depression, pain, and distress in adult patients with cancer, and the outcomes of anxiety and pain in pediatric patients with cancer.

## 3. Results

### 3.1. Characteristics of included trials

#### 5.1.1. Study description

Table 1 lists the characteristics, settings, and interventions. The 15 included trials had a mean age range of 42.69–59.03 years for adult patients and 10.2–15.28 years for pediatric patients. Ten trials involved adult patients (Chirico et al., 2020; Fabi et al., 2022; Ioannou et al., 2022; Mohammad and Ahmad, 2019; Oyama et al., 2000; Schneider et al., 2004; Schneider and Hood, 2007; Verzwyvelt et al., 2021; Zhang et al., 2023; Zhang et al., 2022b), while five involved pediatric patients (Erdős and Horváth, 2023; Gerçeker et al., 2021; Gershon et al., 2004; Wong et al., 2021, 2022). A total of 804 participants, including 547 adult and 257 pediatric patients, were included in this review. All participants were diagnosed with cancer and actively receiving chemotherapy. Most adult cancer patients were women with breast cancer, while others had ovarian, lung, colon, and other solid types of cancer. The number of pediatric patients by sex was almost equal, but the type of cancer was not specified.

Our initial search yielded 993 articles, of which 202 were duplicates. In addition, we identified 14 articles from citation search and Google Scholar. After screening the titles, abstracts and full-text screening, finally, 15 articles that met the inclusion criteria were included in this review. Of the 15 articles, 12 were included in the quantitative synthesis (meta – analysis). Detail PRISMA flow diagram was shown in Fig. 1.

In each study, various brands of VR tools were used, all of which had a head-mounted display model and characteristics identical to those of IVR. Adult patients with cancer were offered virtual environments in the form of exploring natural landscapes such as the sea, parks, forests, beaches, waterfalls, and mountains (Chirico et al., 2020; Fabi et al., 2022; Ioannou et al., 2022; Mohammad and Ahmad, 2019; Oyama et al., 2000; Schneider et al., 2004; Schneider and Hood, 2007; Verzwyvelt et al., 2021; Zhang et al., 2023; Zhang et al., 2022b), walking through an art museum (Schneider et al., 2004; Schneider and Hood, 2007), and watching a concert and yoga sessions (Fabi et al., 2022). Pediatric patients with cancer were provided with virtual environment in the form of games (Erdős and Horváth, 2023), outdoor activities (Gerçeker et al.,

**Table 1**
Characteristics of the included trials.

| Author [Year] | Inclusion criteria | Study design | Sample size, N (% of women) | Age (years) (mean ± SD) | Setting | Intervention |
|---|---|---|---|---|---|---|
| Adult patient (≥18 years) | | | | | | |
| Chirico et al. [2020] | Patients with breast cancer aged between 18 and 70 years, receiving IV chemotherapy | RCT | EG1: 30 (100%)<br>EG2: 30 (100%)<br>CG: 30 (100%) | EG1: 55.18 ± 5.7<br>EG2: 55.7 ± 5.26<br>CG: 56.2 ± 6.79 | During chemotherapy | EG1:<br>D: Head-mounted glasses (Vuzix Wrap 1200 VR) with a head motion tracking system<br>VE: Relaxing landscapes, participants explored an island, by walking through a forest, observing different animals, climbing a mountain, and swimming in the sea.<br>T, F: 20 min, once<br>EG2:<br>Patients listened to 20-min relaxing music pre-taped by an expert music therapist<br>D: Mp3 reader and headphones<br>T, F: 20 min, once<br>CG: Chemotherapy usual care |
| Fabi et al. [2022] | Histological diagnosis of stage I to III breast or ovarian cancer, aged ≥18 years, ECOG 0 to 2, life expectancy >12 months | RCT | EG: 22 (100%)<br>CG: 22 (100%) | EG: BC 50 (57–71); GC 50 (36–61)<br>CG: BC 50 (39–69); GC 52 (51–62) | During chemotherapy | EG:<br>D: VR headset Oculus Go<br>VE: Relaxing and engaging content, such as concerts, walks in the European capitals, mountain nature trails, isolated and fascinating places, pristine, exotic beaches, and yoga sessions<br>T, F: 10 min, once<br>CG: Standard care |
| Ioannou et al. [2022] | Patients with histopathological diagnosis of cancer, aged ≥18 years, and on active chemotherapy treatment | Crossover RCT | 50 (42.0%) | 57 ± 15.5 | During chemotherapy | EG:<br>D: VIVE VR Headset, a head-mounted display (HMD)<br>VE: A sunny environment with a waterfall flowing from snowy mountains into a lake, Sound of falling water and bird sounds were incorporated over a relaxing soundtrack<br>T, F: 20 min, once<br>CG: Standard care |
| Mohammad & Ahmad [2019] | Female patients diagnosed as having breast cancer, aged between 18 and 70 years | RCT | EG: 40 (100%)<br>CG: 40 (100%) | 30–70 (51.99 ± 10.34) | In chemotherapy phases | EG:<br>D: Head-mounted display with headphones; not specific described<br>VE: deep-sea diving "Ocean Rift" and sitting on the beach with the "Happy Place" track<br>T, F: 15 min, once<br>CG: Standard care |
| Oyama et al. [2000] | Patients with cancer receiving chemotherapy, aged 18–70 years | RCT | EG: 15 (80%)<br>CG: 15 (80%) | 18–70 (53.5) | During chemotherapy | EG:<br>D: Three-screen liquid-crystal display (LCD) with a 3D system, headphones, and speakers<br>VE: Three virtual worlds consisting of lake, forest, and country town.<br>T, F: 20 min, 1 time<br>CG: Chemotherapy usual care |
| Schneider et al. [2004] | Women with breast cancer who were scheduled to receive IV chemotherapy, aged 18–55 years | Crossover RCT | 20 (100%) | 27–55 (42.6 ± 7.9) | During chemotherapy | EG:<br>D: Sony PC Glasstron PLMS700 (head-mounted display)<br>VE: Deep-sea diving, walking through an art museum, or solving a mystery<br>T, F: 45–90 min, once<br>CG: Chemotherapy usual care |
| Schneider et al. [2007] | First diagnosis of breast, colon or lung cancer, aged ≥18 years, receiving IV chemotherapy | Crossover RCT | 123 (77%) | 32–78 (53.97 ± 10.89) | During chemotherapy | EG:<br>D: VR HMD; i-Glasses SVGA head-mounted Display, i-O display systems<br>VE: deep-sea diving, walking through an art museum, exploring ancient worlds, and solving a mystery<br>T, F: 45–90 min, once<br>CG: Chemotherapy usual care |
| Verzwyvelt et al. [2021] | Diagnosed as having stage 0 to IV solid cancer, aged ≥18 years | Crossover RCT | 33 (75.8%) | 26–84 (59.03 ± 13.2) | During chemotherapy | EG:<br>D: Oculus Quest Head-Mounted Display (HMD). The VR HMD is a stand-alone device with built-in tracking and headphones (to allow audio) and has head tracking and a separate hand-controller<br>VE: The Nature Treks software. Patients can explore tropical beaches, underwater oceans, and even watch the stars. |

(*continued on next page*)

**Table 1** (*continued*)

| Author [Year] | Inclusion criteria | Study design | Sample size, N (% of women) | Age (years) (mean ± SD) | Setting | Intervention |
|---|---|---|---|---|---|---|
| | | | | | | T, F: 53.3 min, once<br>CG: Standard care |
| Zhang et al. (2022b) | Patients with pathologically diagnosed breast cancer, aged between 18 and 70 years | RCT | EG: 38 (100%)<br>CG: 39 (100%) | EG: 52.29 ± 7.68<br>CG: 51.03 ± 7.97 | Post chemotherapy | EG:<br>D: Oculus Go VR headset<br>VE: A stereoscopic visual scene (Tuscany Garden)<br>T, F: 30 min, 6 times (3 months)<br>CG: Standard care |
| Zhang et al. [2023] | Newly diagnosed with acute leukemia, aged ≥18 years, Karnofsky performance >60 | RCT | EG: 30 (56.7%)<br>CG: 30 (53.3%) | EG: 33.5 ± 11.1<br>CG: 35.3 ± 10.6 | During chemotherapy | EG:<br>D: A head-mounted display (PRO6DOF model, IQiyi Intelligent Tech)<br>VE: A 360-degree panoramic (e.g., forest, beach) and meditation music<br>T, F: 20 min/session, 14 days<br>CG: Usual care |
| Pediatric patients (≤18 years) | | | | | | |
| Erdős & Horváth [2023] | Children with cancer aged 10–18 years receiving chemotherapy | Crossover RCT | 29 (27.5%) | 15.28 ± 2.44 | During chemotherapy | EG:<br>D: the Samsung Gear VR (with Samsung Galaxy S7 Edge) and the Oculus Go<br>VE: VR game Night sky (Digital games)<br>T: 30 min, once<br>CG: Standard care |
| Gerçeker et al. [2021] | Children or adolescents with cancer aged 6–17 years undergoing Huber needle insertion for routine chemotherapy | RCT | EG: 21 (38.1%)<br>CG: 21 (38.1%) | NA | Pre chemotherapy | EG:<br>D: Samsung Gear Oculus headset, connected to the Samsung Galaxy S7 Edge<br>VE: Swimming with marine animals underwater (Ocean Rift), riding a rollercoaster (Rilix VR), and exploring the forest through the eyes of woodland species (in the eyes of animal)<br>T: 10 min, once<br>CG: Standard care |
| Gershon et al. [2004] | Children with cancer, aged 7–19 years, receiving a port access for chemotherapy | Pilot RCT | EG: 22 (NA)<br>CG1: 22 (NA)<br>CG2: 15 (NA) | 12.7 ± NA | Pre chemotherapy | EG:<br>D: A head-mounted display with stereo earphones<br>VE: Virtual Gorilla program, created as an educational tool for children visiting the gorilla habitat at Zoo Atlanta<br>T: 5–10 min, once<br>CG: Standard care |
| Wong et al. [2021] | Pediatric patients with cancer, aged 6–17 years | RCT | EG: 54 (44.4%)<br>CG: 54 (40.7%) | EG: 10.5 ± 3.8<br>CG: 10.2 ± 3.5 | In chemotherapy phases | EG:<br>D: Google cardboard goggle fitted to Apple and Samsung smartphones<br>VE: VR cartoons over VR museum or VR water worlds and "Minion" movies<br>T: 10 min, once<br>CG: Standard care |
| Wong et al. [2022] | Children with cancer, aged 6–12 years, who receiving first chemotherapy | Exploratory – RCT | EG: 9 (44.4%)<br>CG: 10 (30%) | EG: 10.33 ± 1.50<br>CG: 9.11 ± 1.60 | During chemotherapy | EG:<br>D: Google Cardboard goggles<br>VE: Minion mini movies, Doraemon mini movie, and a spider journey 3D cartoon<br>T: 30 min (1st session), 5 min (2nd session), 5 min (3rd session),3 time<br>CG: Standard care |

RCT: Randomized controlled trial; CG: Control Group; EG: Experiment Group; D: Device; VE: Virtual Environment; T: Time duration; F: Frequency; BC: Breast Cancer; GC: Gynecological Cancer.

2021; Gershon et al., 2004; Wong et al., 2021), and cartoon movies (Wong et al., 2021, 2022). In all trials, the VR device was used for durations ranging from 5 to 90 min, with a frequency of use of once per chemotherapy session or in accordance with standard care such as watching TV, reading a newspaper/book, listening to music, or receiving only standard information and doing nothing.

### 3.1.2. *Methodological risk of bias*

According to the results of RoB 2.0, eight trials had an overall low risk of bias, while only one trial had a high risk of bias (Erdős and Horváth, 2023). In the domain of bias arising from the randomization process, nine trials had a low risk of bias. In the domain of bias due to deviations from the intended intervention, two trials that did not provide information related to this domain were determined as having some concerns. In the domain of bias due to missing outcome data, all the included trials had a low risk of bias. In the domain of bias in measurement of the outcome, only one trial that used a measurement instrument with poor validity had a high risk of bias. In the domain of bias in the selection of reported results, all the included trials had a low risk of bias. All trials reported expected outcomes in detail (Table 2). We acknowledge that performance bias (blinding) is difficult to apply to VR research due to the nature of the intervention.

## 3.2. *Effect of IVR intervention in adult cancer patients*

Of the 10 trials, six examined anxiety (Chirico et al., 2020; Fabi et al., 2022; Ioannou et al., 2022; Mohammad and Ahmad, 2019; Schneider and Hood, 2007; Zhang et al., 2023). Four trials examined depression

**Table 2**
Assessment of Methodological Quality of Included trials (RCT evaluated by RoB 2.0).

| Author [year] | Bias arising from the randomization process | Bias due to deviations from intended interventions | Bias due to missing outcome data | Bias in measurement of the outcome | Bias in selection of the reported results | Overall risk of bias |
|---|---|---|---|---|---|---|
| Chirico et al. (2020) | Some concerns | Some concerns | Low | Low | Low | Some concerns |
| Erdős and Horváth (2023) | Low | Some concerns | Low | High | Low | High |
| Fabi et al. (2022) | Low | Low | Low | Low | Low | Low |
| Gerceker et al. (2021) | Low | Low | Low | Low | Low | Low |
| Gershon et al. (2004) | Some concerns | Low | Low | Low | Low | Some concerns |
| Ioannou et al. (2022) | Low | Low | Low | Low | Low | Low |
| Mohammad and Ahmad (2019) | Low | Low | Low | Low | Low | Low |
| Oyama et al. (2000) | Some concerns | Low | Low | Low | Low | Some concerns |
| Schneider et al. (2004) | Some concerns | Low | Low | Low | Low | Some concerns |
| Schneider et al. (2007) | Low | Low | Low | Low | Low | Low |
| Wong et al. (2022) | Low | Low | Low | Low | Low | Low |
| Wong et al. (2021) | Low | Low | Low | Low | Low | Low |
| Verzwyvelt et al. (2021) | Some concerns | Low | Low | Low | Low | Some concerns |
| Zhang et al. (2022a, b) | Low | Low | Low | Low | Low | Low |
| Zhang et al. (2023) | Low | Low | Low | Low | Low | Low |

(Chirico et al., 2016; Fabi et al., 2022; Ioannou et al., 2022; Zhang et al., 2022b), and three trials examined distress. Five trials examined changes in fatigue and three trials examined pain. Two trials examined systolic blood pressure and one trial examined heart rate.

### 3.2.1. *Anxiety*

This study examined changes in anxiety in adult patients from baseline to immediately after VR (6 trials) and from baseline to 48 h after VR (2 trials). One trial could not be pooled due to insufficient data (Schneider et al., 2004). The State Anxiety Inventory (SAI), Self – rating Anxiety Scale (SAS), and anxiety subscale of the Profile of Mood States – Brief (POMS-B) instrument were used to assess anxiety. The pooled results revealed a significant difference in anxiety between the intervention and standard care groups (SMD = −1.73, 95% CI = −2.59 to −0.86, $P < 0.0001$). In the subgroup analysis, the IVR group had significantly lower anxiety than did the standard care group immediately after VR (SMD = −2.13, 95% CI = −3.38 to −0.89, P = 0.0008). However, there were no significant differences in anxiety between the IVR and standard care group 48 h after VR (SMD = 0.59, 95% CI = 1.64 to 0.45, P = 0.27). High heterogeneity was observed between the trials using the random-effects model (Q statistic = 149.27, $I^2$ = 97%, $P < 0.00001$, Fig. 2A).

### 3.2.2. *Depression*

The studies examined changes in depression in adult patients from baseline to immediately after VR and from baseline to 48 h after VR. Depression was assessed using the depression subscales in the Profile of Mood States (POMS), Self – Rating Depression Scale (SDS), and Hospital Anxiety and Depression Scale (HADS). The pooled results revealed a significant difference in depression between the intervention and standard care groups (SMD = −2.21, 95% CI = −3.97 to −0.45, P = 0.01). In the subgroup analysis, the IVR group had significantly lower depression than the standard care group immediately after intervention (SMD = −3.04, 95% CI = −5.46 to −0.62, P = 0.01). However, Fabi et al. (2022) reported no difference in depression between the groups 48 h after intervention (SMD <0.001, 95% CI = −0.59 to 0.59). High heterogeneity was observed between the trials using the random-effects model (Q statistic = 86.74, $I^2$ = 97%, $P < 0.00001$, Fig. 2B).

### 3.2.3. *Distress*

Distress in adult patients was assessed using several scales, including the Adapted Symptom Distress Scale – 2 (ASDS-2), Distress Screening Assessment Tool (DSAT), Distress Thermometer (DT), and Symptom Distress Scale (SDS). The pooled results showed no significant difference in distress between the IVR and standard care groups (SMD = −0.63, 95% CI = −1.77 to 0.52, P = 0.28). High heterogeneity was observed between the trials using the random-effects model (Q statistic = 43.09, $I^2$ = 95%, $P < 0.00001$, Fig. 2C). One trial was not included in the forest plot analysis because the data could not be pooled, but no significant changes in distress were observed between groups (Schneider et al., 2004).

### 3.2.4. *Fatigue*

Fatigue was assessed using several scales, including the Piper Fatigue Scale (PFS), fatigue subscale in the POMS, Cancer Fatigue Scale (CFS), and fatigue subscale in the European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Core 30 (EORTC QLQ – 30). Subgroup analysis revealed that the IVR group had significantly lower fatigue than did the standard care group immediately after intervention (SMD = −3.40, 95% CI = −5.54 to −1.26, P = 0.002). However, no significant difference in fatigue was observed between the IVR and standard care groups 48 h after intervention (SMD = −0.11, 95% CI = −0.13 to 0.35, P = 0.38). High heterogeneity was observed between trials using the random effects model (Q statistic = 158.60, $I^2$ = 98%, $P < 0.00001$). Two trials reported a significant difference in fatigue scores between the IVR and standard care groups ($P < 0.05$, Fig. 2D). One trial observed significant changes in fatigue from baseline to 48 h after intervention (Oyama et al., 2000), while another trial observed significant changes immediately after chemotherapy (Schneider et al., 2004). These trials could not be pooled into the forest plot analysis due to insufficient data.

### 3.2.5. *Pain*

We examined changes in pain in adult patients with cancer from baseline to immediately after IVR (2 trials) and from baseline to 48 h after IVR (1 trials). Pain was assessed using the Visual Analog Scale (VAS), Numeric Pain Scale (NPS), and pain subscale in the EORTC QLQ – 30 were used to assess pain. The pooled results showed no significant

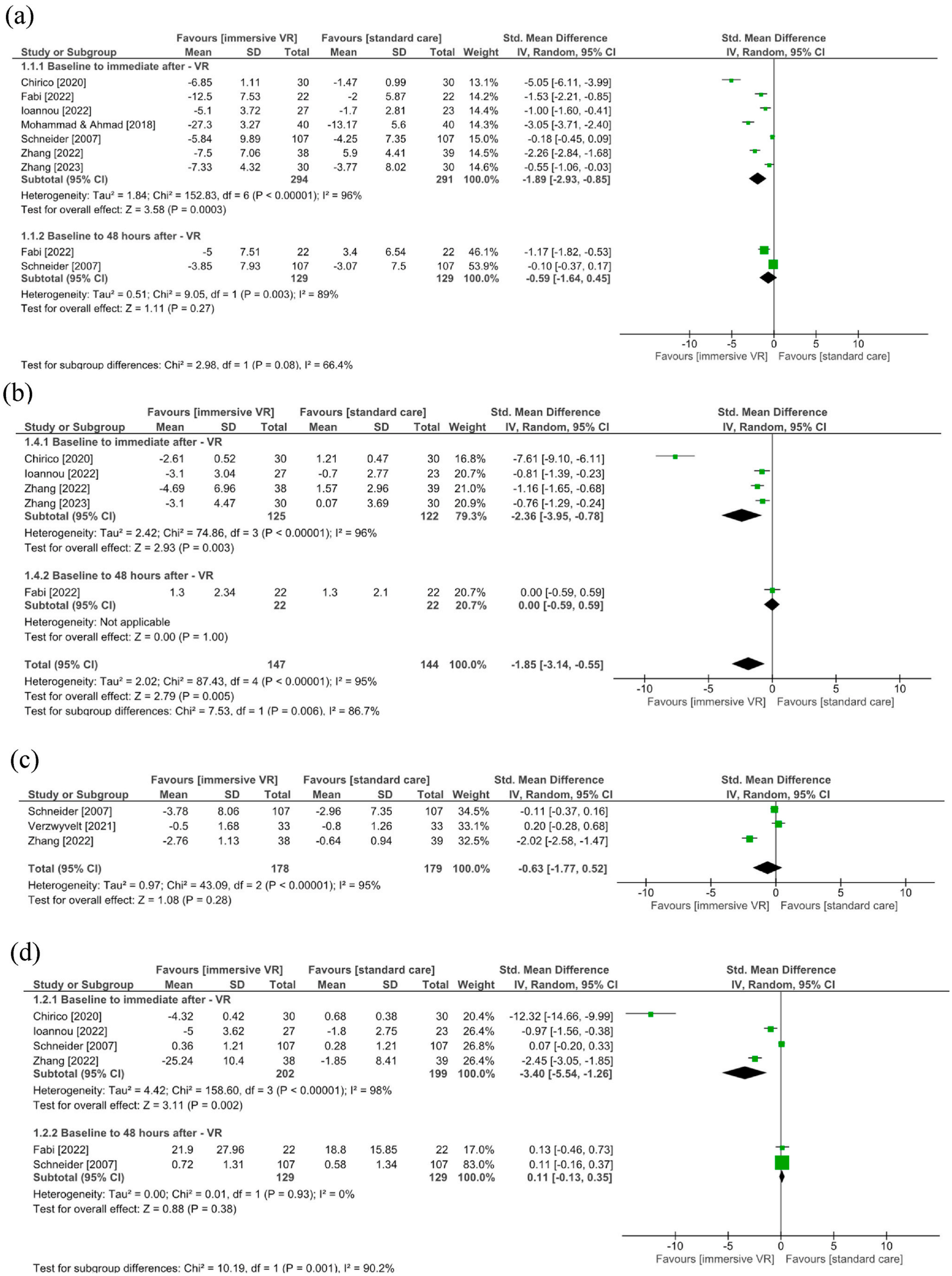


**Fig. 2.** Forest plot comparing immersive virtual reality versus standard care; adult cancer patients: (a) anxiety; (b) depression; (c) distress; (d) fatigue; (e) pain; (f) systolic blood pressure.

difference in pain between groups (SMD = −0.70, 95% CI = −2.66 to 1.27). Subgroup analysis revealed no significant difference in pain between the IVR and standard care groups after VR (SMD = −1.14, 95% CI = −4.34 to 2.06, P = 0.49). High heterogeneity was observed between the trials using the random-effects model (Q statistic = 72.22, $I^2$ = 97%, P < 0.00001, Fig. 2E).

*3.2.6. Blood pressure and heart rate*

The mean difference in systolic blood pressure between the IVR and standard care groups was significant (MD = −3.54, 95% CI = −6.67 to −0.40, P = 0.03, Fig. 2F). Diastolic blood pressure was reported to change significantly in the study by Ioannou et al. (2022), but not in the study by Verzwyvelt et al. (2021). Heart rate change was reported to be significant in the study by Ioannou et al. (2022) after the VR intervention, but not in the study by Verzwyvelt et al. (2021). Diastolic blood pressure and heart rate data were not included in the pooled analysis due to insufficient data.

### 3.3. Effect of IVR intervention in pediatric cancer patients

Of the 5 trials, three trials examining examined anxiety (Gershon et al., 2004; Wong et al., 2021, 2022), two examined pain (Gerçeker et al., 2021; Wong et al., 2021), and three examined heart rate (Gershon et al., 2004; Wong et al., 2021, 2022).

*3.3.1. Anxiety*

Anxiety was assessed using several scales, including the State Anxiety Inventory (SAI), the Children's Anxiety Meter-State (CAM-S), and the State Anxiety Scale for Children (CSAS – C). The pooled results revealed a significant difference in anxiety between the IVR and standard care groups (SMD = −1.18, 95% CI = −1.77 to −0.59, P < 0.0001). Low heterogeneity was observed across the trials (Q statistic = 4.66, $I^2$ = 57%, P = 0.10, Fig. 3A). Although Gershon et al. (2004) and Erdős and Horváth (2023) also reported anxiety outcomes, these trials could not be included in the forest plot analysis due to insufficient data. However, the findings of Gershon et al. (2004) are in line with the pooled results, where a significant difference in anxiety was noted between the groups. On the other hand, Erdos et al. (2022) reported no significant differences in anxiety between the intervention and standard care groups.

*3.3.2. Pain*

The pain was assessed using the Visual Analog Scale (VAS) and the Wong–Baker Faces Pain Rating Scale (WBSFPRS). The pooled results revealed a significant difference in pain between the IVR and standard care groups (SMD = −1.17, 95% CI = −1.84 to −0.50, P = 0.0006). Moderate heterogeneity was observed across the trials (Q statistic = 2.85, $I^2$ = 65%, P = 0.09, Fig. 3B). Two trials could not be pooled into the forest plot analysis due to data unavailability. The findings of Gershon et al. (2004) are in line with those of the forest plot analysis, where a significant difference in pain was observed between the groups. However, another trial showed no significant differences (Erdős and Horváth, 2023).

*3.3.3. Heart rate*

Three trials examining heart rate in pediatric patients were included in the meta-analysis. The mean heart rate difference was 0.33 (95% CI = −3.34 to 3.99; P = 0.86, Fig. 3C), indicating no significant variation in heart rate. Low heterogeneity was observed across trials ($I^2$ = 20%).

### 3.4. GRADE certainty assessment

The overall quality of evidence was graded as very low for adult patients with cancer and low for pediatric patients with cancer, according to the GRADE certainty assessment (Table 3). For the outcomes measured in both adult (anxiety, depression, fatigue, pain, and distress) and pediatric (anxiety and pain) cancer populations, the risk of bias was downgraded due to selection bias and/or performance bias inherent in VR interventions. Inconsistency was downgraded due to high heterogeneity, while indirectness was downgraded because virtual reality interventions varied in components, frequency, approach, and duration across different trials. Imprecision was downgraded because most trials had fewer than 50 individuals per group, resulting in broad confidence intervals. No publication bias was detected due to the small number of trials.

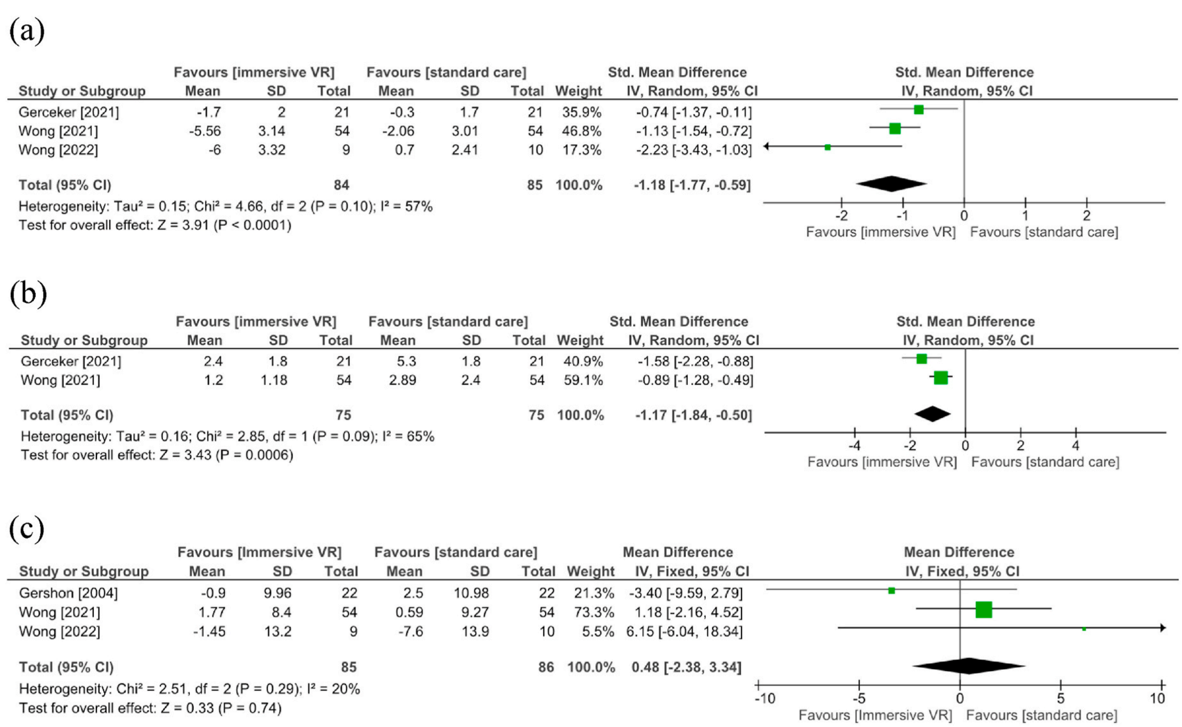


**Fig. 3.** Forest plot comparing immersive virtual reality versus standard care; pediatric cancer patients: (a) anxiety; (b) pain; (c) heart rate.

**Table 3**
GRADE summary of evidence.

| No. of studies and design | Certainty assessment | | | | | No. of patients | | Effect | Certainty | Importance |
|---|---|---|---|---|---|---|---|---|---|---|
| | Risk of bias | Inconsistency | Indirectness | Imprecision | Other bias | Intervention group | Control group | SMD [95% CI] | | |
| Anxiety (adult) (follow-up: mean 1 days) | | | | | | | | | | |
| 5 RCT<br>2 Crossover | Not Serious[a] | Serious[b] | Not Serious[c] | Serious[d] | None | 294 | 291 | −1.89 [−2.93 to −0.85] | ⊕⊕◯◯ LOW | IMPORTANT |
| Depression (adult) (follow-up: mean 1 days) | | | | | | | | | | |
| 3 RCT<br>1 Crossover | Not Serious[a] | Serious[b] | Not Serious[c] | Serious[d] | None | 125 | 122 | −2.36 [−3.95 to −0.78] | ⊕⊕◯◯ LOW | IMPORTANT |
| Fatigue (adult) (follow-up: mean 1 days) | | | | | | | | | | |
| 2 RCT<br>2 Crossover | Not Serious[a] | Serious[b] | Not Serious[c] | Serious[d] | None | 202 | 199 | 3.40 [−5.54 to −1.26] | ⊕⊕◯◯ LOW | IMPORTANT |
| Pain (adult) (follow-up: mean 1 days) | | | | | | | | | | |
| 1 RCT<br>1 Crossover | Serious[a] | Serious[b] | Not Serious[c] | Very Serious[d] | None | 73 | 73 | −1.14 [−4.34 to 2.06] | ⊕◯◯◯VERY LOW | IMPORTANT |
| Anxiety (paediatric) (follow-up: mean 1 days) | | | | | | | | | | |
| 3 RCT | Not Serious[a] | Not Serious | Not Serious[c] | Very Serious[d] | None | 84 | 85 | −1.18 [−1.77 to −0.59] | ⊕⊕◯◯ LOW | IMPORTANT |
| Pain (paediatric) (follow-up: mean 1 days) | | | | | | | | | | |
| 2 RCT | Not Serious[a] | Not Serious | Not Serious[c] | Very Serious[d] | None | 75 | 75 | −1.17 [−1.84 to −0.50] | ⊕⊕◯◯ LOW | IMPORTANT |
| Heart rate (paediatric) (follow-up: mean 1 days) | | | | | | | | | | |
| 3 RCT | Not Serious[a] | Not Serious | Not Serious[c] | Very Serious[d] | None | 85 | 86 | 0.48 [-2.38 to 3.34] | ⊕⊕◯◯ LOW | IMPORTANT |

CI: confidence interval; MD: mean difference; SMD: standardized mean difference; RCT = Randomized controlled trials.
Explanations.
[a] Most trials had no limitation, although it is impossible avoid selection/performance bias due to the nature of VR interventions but confidence in result likely not affected.
[b] High heterogeneity.
[c] The design of virtual reality interventions varies in terms of device, frequency, approach, and duration across different trials but most trials had similar concept of VR environment and inclusion criteria.
[d] Most trials had a sample size of fewer than 50 individuals per group, resulting in broad confidence intervals.

## 4. Discussion

A systematic review and meta-analysis was conducted to assess the efficacy of IVR in alleviating symptoms among cancer patients receiving chemotherapy. Specifically, we evaluated anxiety, fatigue, distress, depression, pain, heart rate, and blood pressure in adult patients and pain, anxiety, and heart rate in pediatric patients. In adult patients, we found that IVR group had significantly lower levels of anxiety, depression, and fatigue compared to standard care group after the intervention. Additionally, we noted a significant change in systolic blood pressure between the groups. No notable differences were found between the IVR and standard care groups in terms of distress, pain, and heart rate. However, when examining pediatric patients specifically, the IVR group exhibited significantly reduced levels of pain and anxiety following the intervention when compared to the standard care group. Nevertheless, there was no statistically significant difference in heart rate between two groups.

Our findings indicate the effectiveness of immersive VR and provide an update to the results of a previous meta-analysis that reported inconclusive findings. The previous trial included fewer trials (Rutkowski et al., 2021), focused only on patients with breast cancer in the rehabilitation setting (Bu et al., 2022; Zhang et al., 2022a), and evaluated all types of pain outcomes in adult and pediatric/adolescents with all diseases (Huang et al., 2022). We comprehensively examined not only psychological and physical, but also physiological outcomes (blood pressure and heart rate) in adult and pediatric cancer patients. This is because most studies on VR in cancer patients evaluating self-reported measures have provided inconsistent results (Ahmad et al., 2020; Chirico et al., 2016; Rutkowski et al., 2021). Therefore, the use of these physiological measurements can provide more objective findings.

Advancement of VR devices is continuously ongoing, and the characteristics of VR devices and the type of virtual environment may affect the effectiveness of VR in cancer patients receiving chemotherapy (Rutkowski et al., 2021). Although four trials included in this study reported promising results, they used VR devices that are outdated, complex in application, and not portable (Gershon et al., 2004; Oyama et al., 2000; Schneider et al., 2004; Schneider and Hood, 2007). In contrast, the trials published in 2010 used a more sophisticated VR model with a virtual environment that appeared more real. However, this is not important as long as the virtual content displayed on the VR device enables users to feel like they are immersed and interact with the virtual world in addition to the actual environment (Javaid and Haleem, 2020; Mazurek et al., 2019). The immersive characteristics of VR can completely replace users' visual and sensory senses (Coburn et al., 2017). However, VR interventions can vary widely in three core aspects: the type of equipment used, the content and nature of the virtual world, and the level of engagement the user might have (Lambert et al., 2020). Overall, the desired VR effect is to create an immersive experience where the user is placed in a simulated environment that looks and feels just as engaging as the real world (Chan et al., 2021).

Our results revealed that IVR reduced anxiety, fatigue, and depression and controlled systolic blood pressure in adult patients with cancer during chemotherapy. Although these results differ from those of previous studies (Chow et al., 2021; Rutkowski et al., 2021) in which the included trials did not demonstrate significant changes in anxiety, pain, and fatigue between the intervention and standard care groups, the current findings provide an update with more promising results because we included more trials. In pediatric cancer patients, VR relieved pain and reduced anxiety during chemotherapy. These results are consistent with two meta-analyses, who indicated that VR is effective as distraction therapy in pediatric patients with cancer receiving chemotherapy and surgery (Bukola and Paula, 2017; Cheng et al., 2022). In terms of brain activity, exposure to the VR content causes a decrease in beta wave activity and results in the balancing of alpha brain waves in the anterior cingulate cortex and midline (Tarrant et al., 2018; Wang et al., 2022b). The anterior cingulate cortex and midline regulate attention, anxiety, and mood. These changes in psychological responses during VR intervention can be best measured using physiological monitoring techniques (e.g., blood pressure, pulse rate, respiratory rate, oxygen saturation, and hormone levels (Berto, 2014; Lim et al., 2022). Therefore, nurses or oncology teams should consider this VR intervention approach in cancer patients as a technology-based nonpharmacological therapy not only in chemotherapy settings, but also in other clinical settings.

The results of this study are promising and may have potential implications for future cancer treatment. However, there are some limitations that should be considered. First, only 15 trials were included involving 607 adult patients and 257 pediatric cancer patients. Although the sample is larger than that included in a previous meta-analysis (Burrai et al., 2023; Rutkowski et al., 2021), particularly in the adult population, and we also tested physiological outcomes in the pediatric population, which the previous meta-analysis did not (Czech et al., 2023). Controlling for heterogeneity in these findings was challenging due to the majority of studies used multiple outcome measurement tools, involved different cancer types, employed varied VR scenarios and devices, and included different intervention durations. Furthermore, the inherent characteristics of VR made it challenging to blind the majority of the studies, which affected the methodological rigor. Given the indeterminate quality of the evidence and the suggested results, these findings should be carefully considered and cautiously applied in future research. Consider larger sample sizes, extended observation, and follow-up intervals, testing the feasibility of VR acceptability, evaluating VR side effects, and optimizing VR content and tools in future studies. Furthermore, a consensus is needed to standardize VR implementation (duration, tools, content, user criteria) in oncology service settings.

## 5. Conclusion

This systematic review and meta-analysis demonstrated the efficacy of immersive virtual reality (IVR) in reducing psychological problems (such as anxiety and depression), physical symptoms (such as fatigue), and physiological symptoms (such as pulse and blood pressure) in cancer patients undergoing chemotherapy. IVR maybe considered by the oncology team as an alternative nonpharmacological intervention cancer patient receiving chemotherapy. Nonetheless, further well-constructed RCTs are necessary to thoroughly assess the use of IVR in terms of content, devices, suitable settings, and effective duration to ensure comfort and satisfaction to its users.

## CRediT authorship contribution statement

**Made Satya Nugraha Gautama:** Conceptualization, Methodology, Software, Data curation, Writing – original draft, Writing – review & editing. **Tsai-Wei Huang:** Conceptualization, Methodology, Software, Data curation, Writing – original draft, Writing – review & editing, Supervision. **Haryani Haryani:** Methodology, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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